ATM MANAGEMENT SYSTEM

C++ Source Code and Program Output

1. C++ Source Code

```
#include <iostream>
using namespace std;

class ATM {
private:
    int balance;

    bool authenticate(int pin) {
        return pin == 1234;
    }

public:
    ATM() {
        balance = 50000;
    }

    void checkBalance(int pin) {
        if (authenticate(pin)) {
            cout << "Your balance is: "
                 << balance << " FCFA" << endl;
        } else {
            cout << "Incorrect PIN!" << endl;
        }
    }

    void withdraw(int pin, int amount) {
        if (authenticate(pin)) {
            if (amount > 0 && amount <= balance) {
                balance -= amount;

                cout << "Withdrawal successful!" << endl;
                cout << "Remaining balance: "
                     << balance << " FCFA" << endl;
            } else {
                cout << "Insufficient balance or invalid amount!"
                     << endl;
            }
        } else {
            cout << "Incorrect PIN!" << endl;
        }
    }
};

int main() {
    ATM atm;
    int pin, choice, amount;

    cout << "===== WELCOME TO ATM =====" << endl;
    cout << "Enter your PIN: ";
    cin >> pin;

    do {
        cout << "\n===== ATM MENU =====" << endl;
        cout << "1. Check Balance" << endl;
        cout << "2. Withdraw Money" << endl;
        cout << "3. Exit" << endl;

        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                atm.checkBalance(pin);
                break;

            case 2:
                cout << "Enter amount to withdraw: ";
                cin >> amount;
                atm.withdraw(pin, amount);
                break;

            case 3:
                cout << "Thank you for using our ATM!" << endl;
```

```
                break;

            default:
                cout << "Invalid choice!" << endl;
        }

    } while (choice != 3);

    return 0;
}
```

2. Program Output

The screenshot below shows the program running successfully: checking the balance, withdrawing 2,000 FCFA, displaying the remaining balance, and exiting the ATM menu.

